

Transport closure: direct sums of weight systems compose exactly

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Abstract

A single functorial lift is one theorem; a transport calculus is a system. This note proves the closure property of the carrier-transport machinery: when two weight systems are combined by direct sum, every layer of the arithmetic passport composes exactly — local Euler coefficients by Cauchy product, global coefficient banks by Dirichlet convolution, prescribed completions by multiplication at the product conductor and concatenated shift list, and self-dual functional equations by multiplication. The abstract composition laws (identity, associativity, faithfulness under composition, base-change degrees multiplying) are compiled separately. Two arithmetic instances tie the calculus to a live eigenform: the tensor-square decomposition $\text{std} \otimes \text{std} = \text{Sym}^2 \oplus \mathbf{1}$, whose bank is the rank-4 Rankin bank $\zeta \star \text{sym2Bank}$, and the plethysm $\text{Sym}^2 \circ \text{Sym}^2 = \text{Sym}^4 \oplus \mathbf{1}$, whose bank is $\zeta \star (\text{Sym}^4\text{-bank})$ — both derived from the closure law, not verified numerically. Everything is unconditional and machine-checked.

1 Setup

A *weight system* of rank d is a family $w = (w_i)_{i \in I}$ of d nonzero complex numbers, one family per prime in the global setting. Its *local Euler coefficient* at level j is the complete homogeneous symmetric function

$$h_j(w) = \sum_{\substack{(l_i) \geq 0 \\ \sum l_i = j}} \prod_{i \in I} w_i^{l_i},$$

the p^j -coefficient of $\prod_i (1 - w_i p^{-s})^{-1}$. Its *global bank* is the multiplicative arithmetic function assembled from the local coefficients over the prime factorization, $c_w(n) = \prod_{p|n} h_{v_p(n)}(w_p)$; formally the shift-by-one of the all-place coefficient sequence (`coefficientArithmetic` of `radialGlobalSatakeCoef`). Its *prescribed completion* at conductor $C > 0$ and shift list (μ_j) is $C^s \prod_j \Gamma_{\mathbb{C}}(s + \mu_j) \cdot \sum_n c_w(n) n^{-s}$.

Two conventions used throughout: banks are compared as arithmetic functions (so “the bank of w ” means the shifted sequence), and a *direct sum* $u \oplus v$ of weight systems is the disjoint union of the two families at every prime.

*Machine-checked companion: `RequestProject/TransportClosure.lean`, 19 named declarations, each at axiom footprint `{propext, Classical.choice, Quot.sound}` or a subset, no `sorry`, no `axiom`, zero `sorryAx`. Every argument below is given in the text; the Lean names locate the corresponding formal statement.

2 The closure laws

Theorem 1 (local closure: the Cauchy law). *For weight systems u of rank d_1 and v of rank d_2 and every level j ,*

$$h_j(u \oplus v) = \sum_{a+b=j} h_a(u) h_b(v).$$

Proof. Fiber the exponent vectors of the union over the total exponent a carried by the u -channels. The slice with u -total a is the product of the u -antidiagonal at level a and the v -antidiagonal at level $j - a$, and the monomial factors accordingly: $\prod_i (u \oplus v)_i^{l_i} = (\prod u_i^{l_i}) (\prod v_i^{l_i})$. Summing slices gives the Cauchy product. The bijection is $l \mapsto (a, l|_u, l|_v)$ with inverse gluing, and both roundtrips are identities coordinate by coordinate. (`TransportClosure.radialLocalEulerCoeff_sumE`) $\square$

Two degenerate values calibrate the law: the local coefficient is invariant under relabeling of channels (`radialLocalEulerCoeff_equiv`), and the unit system — one channel of weight 1 — has $h_j = 1$ for every j (`radialLocalEulerCoeff_unit`), i.e. its bank is ζ .

Theorem 2 (global closure: banks convolve). *The global bank of any weight system is multiplicative (`isMultiplicative_bankArithmetic`), its prime-power values are the local coefficients (`bankArithmetic_prime_pow`), and the bank of a direct sum is the Dirichlet convolution of the banks:*

$$c_{u \oplus v} = c_u \star c_v.$$

Proof. Multiplicativity: at coprime m, n the prime factors split and the factorization exponents restrict, so the defining product splits. Both sides of the convolution identity are then multiplicative, so they agree once they agree at prime powers; there the convolution is the Cauchy product of prime-power values, $(c_u \star c_v)(p^j) = \sum_{a+b=j} c_u(p^a) c_v(p^b)$ (divisors of p^j are the p^a ; `mul_apply_prime_pow`), which is Theorem 1. (`bankArithmetic_sumElim`) $\square$

Theorem 3 (completion and functional-equation closure). *Polynomial Satake dual pairs are closed under direct sum (`sumPair`; exponent the maximum of the two), with coefficient closure (`sumPair_primalCoeff`). On any common point of absolute convergence the prescribed completion of the sum, at the product conductor and the concatenated shift list, is the product of the two prescribed completions:*

$$(C_1 C_2)^s \prod_{\mu \in \mu_1 \uplus \mu_2} \Gamma_{\mathbb{C}}(s + \mu) \cdot L(c_{u \oplus v}, s) = \left(C_1^s \prod_{\mu_1} \Gamma_{\mathbb{C}} \cdot L(c_u, s) \right) \left(C_2^s \prod_{\mu_2} \Gamma_{\mathbb{C}} \cdot L(c_v, s) \right),$$

and self-dual functional equations multiply: if $F(1 - s) = F(s)$ and $G(1 - s) = G(s)$ then $(FG)(1 - s) = (FG)(s)$.

Proof. The conductor factor splits since both conductors are nonnegative reals; the Γ -product splits over the concatenation; and $L(c_u \star c_v, s) = L(c_u, s) L(c_v, s)$ where both factors converge absolutely (Dirichlet-series multiplication, with the shift-by-one bookkeeping handled by the bridge `dirichlet_shift_eq_LSeries`). The functional-equation clause is immediate. (`completedReadout_sumPair`, `FE.mul`) $\square$

Remark 4 (the abstract layer). *Identity, associativity, closure of faithfulness under composition, and the base-change instance $\text{bc}_p \circ \text{bc}_q = \text{bc}_{pq}$ are compiled separately (`CarrierFunctoriality.fithful_comp`,*

`comp_assoc`, `id_comp`, `comp_id`, `bc_comp`, `bc_one`). Together with Theorems 1–3 the transports form a composition calculus in which every layer of the arithmetic passport is preserved — the systems property, as opposed to any single lift.

3 Two live instances at an eigenform

Let f be a level-one holomorphic Hecke eigenform of weight k with Satake parameters α_p , and recall from the Sym^2 companion the clock systems $\text{Sym}^2: (\alpha^2, 1, \alpha^{-2})$ and $\text{Sym}^4: (\alpha^4, \alpha^2, 1, \alpha^{-2}, \alpha^{-4})$, the Sym^2 convolution bank, and the rank-4 Rankin bank $c = \zeta \star \text{sym2Bank}$.

Theorem 5 (tensor-square decomposition). *The direct sum of the Sym^2 clock system and the unit system carries the rank-4 Rankin bank:*

$$c_{\text{Sym}^2 \oplus \mathbf{1}} = \zeta \star \text{sym2Bank} = c.$$

This is $\text{std} \otimes \text{std} = \text{Sym}^2 \oplus \mathbf{1}$ at the bank level: the closure law converts the direct-sum decomposition of the tensor square into the Euler-product factorization $L(f \times f, s) = \zeta(s) L(\text{Sym}^2 f, s)$ of the Sym^2 companion.

Proof. By Theorem 2, $c_{\text{Sym}^2 \oplus \mathbf{1}} = c_{\text{Sym}^2} \star c_{\mathbf{1}}$. The clock bank of Sym^2 is the convolution bank (the coefficient identification of the companion), and the unit bank is ζ (`bankArithmetic_symClock_eq_sym2Bank`, `bankArithmetic_unit_eq_zeta`). (`tensorSquare_bank`.) $\square$

Theorem 6 (plethysm: $\text{Sym}^2 \circ \text{Sym}^2 = \text{Sym}^4 \oplus \mathbf{1}$). *The $\text{Sym}^2 \circ \text{Sym}^2$ weight system — the six pairwise products $w_i w_j$, $i \leq j$, of the Sym^2 clock — decomposes as the Sym^4 clock together with one trivial channel, and consequently its bank is*

$$c_{\text{Sym}^2 \circ \text{Sym}^2} = \zeta \star c_{\text{Sym}^4}.$$

Proof. The six products are $\alpha^4, \alpha^2, 1, 1, \alpha^{-2}, \alpha^{-4}$ (`sym2CompWeight_products`); relabeling by the explicit bijection that sends the second trivial channel to the unit summand exhibits the multiset as $\text{Sym}^4\text{-clock} \sqcup \{1\}$ (`sixSplit`, `sym2CompWeight_decomposition`). Apply relabeling invariance, Theorem 2, the rank-uniform clock-bank identification at $r = 4$, and the unit bank. (`plethysm_bank`.) $\square$

Remark 7 (register and falsifiers). *Everything above is proven and machine-checked; nothing is conditional, and no automorphy is consumed — the two instances read only the compiled Satake seed of the eigenform. The classical content recovered: the Clebsch–Gordan/plethysm decompositions of $\text{GL}(2)$ weight systems and the corresponding Euler-product factorizations, realized as instances of one closure law rather than case-by-case identities. A counterexample would be a pair of weight systems whose union bank differs from the convolution of their banks at some prime power, contradicting Theorem 2; or a level at which the six $\text{Sym}^2 \circ \text{Sym}^2$ products fail the $\text{Sym}^4 \oplus \mathbf{1}$ split, contradicting Theorem 6. Each is formally stated, so a counterexample would require a kernel failure.*

References

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